

Module 4 — Household Utilities

(modules/utilities)

Purpose

Track utility billing periods, physical consumption, and costs across electricity (EDP/Galp/Endesa), gas, and water ("águas" municipal). Compute cost-per-unit, seasonal baselines, and anomaly detection. Distinguish estimated vs. actual readings and reconcile bills to bank transactions. Support Portuguese tariff complexity: time-of-use rates (vazio/cheias/ponta), contracted power (potência contratada), fixed terms, IVA rates, and self-consumption credits.

Actors & User Stories

- *As a household manager*, I upload or manually enter utility bills; the system detects anomalies and alerts me to consumption spikes before I see the bill.
- *As an auditor*, I compare bill totals to transaction history and flag mismatches or missing readings.
- *As an energy analyst*, I see consumption by time-of-use tariff bucket (kWh@vazio, kWh@cheias, kWh@ponta) and track seasonal patterns.
- *As a solar prosumer*, I record self-consumption and injection credits, ensuring net costs reflect grid exchanges only.
- *As a multi-property household*, I roll up utilities across meters in the main residence and summer house, detecting anomalies per address.

Data Model

All monetary fields (`_eur` suffix) are `NUMERIC(10,2)` decimal EUR amounts (e.g. `19.99`), never minor-unit integers; UI displays them pt-PT-formatted (e.g. `19,99€`). **UtilityMeter**: `id`, `entity_id`, `utility_type`(`ELECTRICITY|GAS|WATER`), `provider_merchant_id`, `unit`(`kWh|m3`), `address_id?`, `contracted_power_kw?` (electricity), `tariff_plan_id?`, `tariff_type`(`TIME_OF_USE|FLAT|TIERED`), `start_date`, `end_date`, `is_deleted`, `notes`.

UtilityBill (distinct from `Reading`): `id`, `meter_id`, `billing_period_start`, `billing_period_end`, `is_estimated`, `actual_read_date?`, `estimated_read_date?`, `consumption` `NUMERIC(14,4)`, `unit`, `total_cost_eur`, `fixed_cost_eur` ("termo fixo"), `variable_cost_eur` ("termo variável"), `iva_eur`, `credit_eur` (self-consumption/injections), `document_id`, `transaction_id?`, `processing_status`, `confidence`, `decision_reasons`, `tou_breakdown` `JSONB`{`vazio_kwh`, `cheias_kwh`, `ponta_kwh`, `vazio_cost_eur`, `cheias_cost_eur`, `ponta_cost_eur`}.

UtilityMeterReading (raw physical readings): `id`, `meter_id`, `reading_date`, `consumption_kwh_or_m3`, `is_estimated`, `reading_type`(`METER_PHOTO|UTILITY_INVOICE|SMART_METER_API`), `raw_payload` `JSONB`.

TariffPlan (user-curated master): `id`, `provider_id`, `name`, `tariff_type`, `rate_tiers` `JSONB`[{`bucket`, `rate_eur_per_kwh` (`NUMERIC(8,5)`), `hours_weekday`, `hours_weekend`}], `power_tier_limits?`, `fixed_cost_eur`, `start_date`, `end_date`, `iva_rate_pct` (13 PT).

ProviderContract: `id`, `meter_id`, `provider_merchant_id`, `tariff_plan_id`, `contract_start`, `contract_end`, `contracted_power_kw`.

DailyBaselineConsumption (derived, anomaly detection): `id`, `meter_id`, `month_season`, `daily_avg_consumption`, `daily_avg_cost_eur`, `daily_std_dev`, `reading_count`.

Functional Requirements

- **FR-4.1 Consumption Tracking.** Log billing periods with consumption (kWh, gas kWh/m³, water m³) and cost; distinguish estimated vs. actual. Estimated readings flagged and excluded from forecasts.
- **FR-4.2 Efficiency Analytics.** Compute cost/unit (€/kWh, €/m³), daily average consumption/cost, period-over-period % change; compare against seasonal baseline; support time-of-use per-bucket breakdown.
- **FR-4.3 Anomaly Detection.** Alert on consumption spikes >1.5× seasonal baseline or Z-score >2.5 (configurable, default 2.0). Separate cost-spike alerts. Rules: "Read changed but not recorded," "Missing reads (gap >60 days)," "Estimated consecutive—actual expected."
- **FR-4.4 Bill Reconciliation.** Match bills to bank transactions (M2) by date (±5 days), amount (±5%), provider name; surface link confidence; low-confidence flagged.
- **FR-4.5 Time-of-Use Tariff Breakdown.** For vaziao/cheias/ponta plans, parse per-bucket kWh and cost; if detail unavailable, flag for manual entry; render per-bucket trends.
- **FR-4.6 Contracted Power & Fixed Term.** Record `potência contratada` and "termo fixo"; alert if peaks exceed contracted power; track fixed cost separately.
- **FR-4.7 Self-Consumption & Grid Injection Credits.** Record self-consumption and injection credits (`credit_eur`, `credit_reason` `SOLAR_INJECTION|OTHER`); net cost = total – credit.
- **FR-4.8 Estimated-to-Actual Reconciliation.** When an estimated bill is replaced by an actual read, auto-link, compute true difference, adjust cost if rates changed; preserve both for audit.
- **FR-4.9 Provider & Tariff History.** On provider switch/renewal, create new **ProviderContract** with retroactive tariff context; support switching windows with dual bills.
- **FR-4.10 Multi-Meter Household Rollup.** Aggregate consumption/cost across meters (main + secondary addresses); configurable aggregation; household-level anomaly alerts.

Automation Rules

1. **Estimated Reading Detection.** `is_estimated=true` or bill date ≠ actual read date → PENDING_REVIEW; never used in forecasts/baselines.
2. **Anomaly Scoring.** Z-score = (daily_avg – baseline_mean)/baseline_std vs same month/season; Z > threshold (default 2.0) → alert + ReviewTask.
3. **Cost Anomaly.** total_cost > expected by ±15% with no rate change → alert.
4. **Missing Reads.** Gap between consecutive bills > 60 days → alert.
5. **Consecutive Estimated.** N consecutive estimated bills without actual read → escalate alert.
6. **Time-of-Use Learning.** Track shift between buckets month-over-month; surface behavioral efficiency.
7. **Bill↔Transaction Reconciliation.** Confidence = date_proximity × amount_match × merchant_match; auto-accept ≥0.90 else ReviewTask.

UI / Screens

1. **Meter Manager** — list/add/edit/delete meters (address, type, provider, contracted power, tariff).
2. **Bill Entry & Parsing** — upload PDF or manual entry; auto-parse (pdfplumber/OCR); confirm/edit.

3. **Consumption Trend Charts** — daily/monthly with seasonal overlay + anomaly points; toggle consumption/cost.
4. **Time-of-Use Breakdown** — stacked bar (vazio|cheias|ponta) + cost overlay + recommendations.
5. **Anomaly Alerts Panel** — recent anomalies with severity; click-through; dismiss/confirm as expected.
6. **Multi-Meter Household Rollup** — aggregate + per-address breakdown + comparative flags.
7. **Bill Reconciliation Review** — bill ↔ suggested transaction; confirm/reject.

API Surface

CRUD /meters · POST /utility-bills (upload/ingest, idempotent) · GET /utility-bills (filters: meter, date, is_estimated, status) · PATCH /utility-bills/{id} · POST /utility-bills/{id}/confirm · GET/POST /utility-readings · POST /meter-readings/{id}/link-to-bill · GET /meters/{id}/analytics · CRUD /tariff-plans · POST /tariff-plans/{id}/apply-to-meters · GET /utilities/anomalies · POST /utilities/anomalies/{id}/confirm · GET /utilities/reconciliation-suggestions · POST /utilities/{id}/link-to-transaction · GET /households/{id}/utilities/rollup.

Analytics & KPIs

- Consumption trends (monthly/yearly by meter/type; daily avg; vs prior year/3-year avg).
- Cost evolution (total, €/unit, €/day; period % change).
- Seasonal comparison (winter vs summer).
- Time-of-use profile (% per bucket; higher vaziao% ≈ better).
- Anomaly flags (count, trend, dismissal rate).
- Cost-per-occupant; contracted power utilization; self-consumption ratio.

Edge Cases & Validation

1. **Overlapping billing periods** → flag duplicate/overlap.
2. **Gapped periods** (>60 days) → alert.
3. **Unit consistency** — bill must match meter unit.
4. **Estimated vs actual** — delta >10% → alert.
5. **Negative consumption** — allowed for net metering; flag unusual.
6. **Time-of-use data missing** — `tou_breakdown_available=false`; exclude from analytics; offer manual entry.
7. **Rate changes mid-period** — split accounting or flag.
8. **Provider switch during period** — record both; compute net once actual confirmed.
9. **Meter replacement** — link old→new; prevent false anomaly from discontinuity.

Additional / Enriched Requirements

- **ER-4.1 Degree-Day Normalization.** Normalize heating/cooling consumption to standard HDD/CDD (18°C baseline) so warm winters don't read as drops. Opt-in weather fetch.
- **ER-4.2 Behavioral Alerts & Efficiency Recommendations.** Non-alarm insights (e.g., shift load off ponta hours to save).
- **ER-4.3 IVA Rate Tracking & Tax Reporting.** Track `iva_rate_pct` per bill (13% default PT); export VAT-paid summaries.
- **ER-4.4 Subscription/Fixed-Cost Predictability.** Decompose variable + fixed + credits; predict next bill.

- **ER-4.5 Multi-Tariff Optimization.** Simulate 12-month cost under alternative plans; recommend cheapest.
- **ER-4.6 Per-Meter Address & Location Context.** `address_id` shared with Assets; per-location anomalies.
- **ER-4.7 Estimated Reading Follow-Up Workflow.** Celery Beat reminder before next expected read; escalate if unconfirmed.
- **ER-4.8 Smart Meter Data Ingestion (FUTURE).** Optional API polling (EDP) to populate readings automatically.

Open Questions / Decisions

1. **Anomaly thresholds global or per-meter/season?** → *Per-meter, per-season, configurable (default $Z \geq 2.0$).*
2. **Shared meters across entities?** → *Attributed to bill-payer entity; split via `payer_split` JSONB.*
3. **Degree-day normalization auto or opt-in?** → *Opt-in per meter (privacy; external API optional).*
4. **Link estimated bills to transactions?** → *Yes, provisional (`link_status=PROVISIONAL`); auto-update on actual.*
5. **SLA for overdue read?** → *Alert after 40 days; urgent after 60.*
6. **Self-consumption credits netted or separate?** → *Both: show `total_before_credits`, `credit`, `net`.*

Definition of Done

- ☐ Meter CRUD (address, type, unit, provider, contracted power, tariff).
- ☐ Bill upload/parse (PDF + manual); auto-extraction.
- ☐ `is_estimated` flag; estimated excluded from forecasts.
- ☐ UtilityMeterReading + linking for audit.
- ☐ DailyBaselineConsumption + anomaly detection (Z-score, cost %) configurable.
- ☐ Consumption & cost trend charts per meter/type.
- ☐ Time-of-use breakdown (vazio|cheias|ponta) parsing + visualization.
- ☐ Bill↔Transaction reconciliation (M2); confidence scoring.
- ☐ Anomaly alerts UI; confirm/dismiss.
- ☐ Multi-meter rollup with per-address filtering.
- ☐ TariffPlan master + CRUD; time-of-use tiers definable.
- ☐ ProviderContract; provider-switch handled (no false anomalies).
- ☐ Self-consumption credits; IVA tracking (13% default).
- ☐ API + OpenAPI; tests (Z-score, reconciliation, edge cases).
- ☐ Seed: EDP, Galp, EPAL/"Águas", sample tariffs/bills.
- ☐ ≥80% auto-accept on uploads + anomaly detection; low-confidence → ReviewTask.

Integration Contract

- **Exposes:** `UtilityBill`, `UtilityMeter` as recordable entities; `Link` targets for Banking (M2); anomaly alerts to Alert Engine; consumption/cost to Dashboards (M8); per-entity cost to Household budgeting (M7).
- **Consumes:** `Merchant` (providers), `Entity`, `Transaction` (M2), `Document` (bill PDFs), `Category`, `Tag`.